such as, fertilisers for plants and for making the air inert, oxygen (20.95%) for breathing and carbon dioxide (0.03%) for photosynthesis. Some other gases like argon, neon, helium, krypton, hydrogen, ozone, zenon and methane are also present. Besides, water vapour and dust particles make their presence felt in one way or the other.

Air pollution is the contamination of the indoor or outdoor air by a range of gases and solids that modify its natural characteristics and percentage. Air pollutants can be categorised into primary and secondary pollutants.

A **primary pollutant** is an air pollutant emitted directly from a source. A **secondary pollutant** is not directly emitted as such, but forms when other pollutants (primary pollutants) react in the atmosphere.

Primary Pollutants

i) Oxides of Sulphur

ii) Oxides of Nitrogen

iii) Oxides of Carbon

iv) Particulate Matter

v) Other Primary Pollutants

Secondary Pollutants

i) Ground Level Ozone

ii) Smog

9) Pollution of Water

Water pollution may be defined as alteration in the physical, chemical and biological characteristics of water, which may cause harmful effects in human and aquatic life.



Water pollution

In India, water pollution has been taking place on a large scale and since a long period. Both surface and groundwater bodies are polluted to a great extent. The major causes of water pollution in India are:

i) Urbanisation

ii) Industrial effluents

iii) Sewages

iv) Agricultural runoff and improper agricultural practices

v) Seawater intrusion

vi) Solid wastes

Need for Prevention Measures

Prevention is defined as the activities taken to prevent a natural calamity or potential hazard from having harmful effects on either people or economic assets.

- Prevention planning consists of i) hazard identification, and ii) vulnerability assessment.
- Delayed actions may increase the economic losses.
- For developing countries like India, prevention is perhaps the most critical components in managing disasters.

Nature is emerging as a new weapon of mass destruction, do you agree?

Around 22,000 people have died in India in 10 years until 2017 due to major environmental disasters – Indian Meteorology Department.

In the past two decades (1998-2017) over 5,00,000 people have died due to extreme weather events around the world – stated by Global Climate Risk Index Report Published by Germanwatch (German-based non-profit organisation).

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